CSC212A: Software Development Fundamentals

Lecture 06 Detailed Summary: UML Elements, Relationships & Diagrams | Course Leaders: Ms. Sahana P Shankar & Ms. Supriya M S

1. STRUCTURAL BUILDING BLOCKS (NOUNS OF UML)

UML model elements fall into seven distinct structural types representing static conceptual or physical entities:

Structural Element	Definition & Semantic Role	Notation Rendering
Class	Set of objects sharing identical attributes, operations, and semantics.	3-compartment rectangle
Interface	Collection of operations specifying a service of a class or component.	Circle or named socket
Use Case	Sequence of actions yielding an observable result of value to an actor.	Solid ellipse
Collaboration	Society of roles working together to provide cooperative behavior.	Dashed ellipse
Active Class	Class whose instances initiate concurrent threads of control.	Rectangle with heavy border
Component	Physical, replaceable part of a system conforming to interfaces.	Tabbed rectangle
Node	Computational resource existing at runtime with processing memory.	3D Cube

2. BEHAVIORAL, GROUPING & ANNOTATIONAL ELEMENTS

Non-structural elements provide dynamics, organization, and documentation:

- **Behavioral Things (Verbs):** *Interactions* (messaging passed among objects) and *State Machines* (sequence of states an object transitions through in response to events).
- **Grouping Things:** *Packages* (tabbed folders organizing structural and behavioral elements into conceptual groups).
- **Annotational Things:** *Notes* (dog-eared rectangles rendering comments or constraints).

3. FOUR CORE UML RELATIONSHIPS

Relationship	Semantic Meaning	Visual Notation
Dependency	Semantic connection where a change in an independent element affects the dependent element.	Dashed directed arrow
Association	Structural connection depicting links between object instances across peer classes.	Solid line with role & multiplicity
Generalization	Inheritance connection where specialized child instances inherit structure and behavior from a parent.	Solid line with hollow arrowhead
Realization	Semantic relationship where one classifier guarantees to execute a contract defined by an interface.	Dashed line with hollow arrowhead

4. MULTIPLICITY & CONTAINMENT MECHANICS

Multiplicity Bounds: Expresses allowable instance counts on associations: 1 (exactly one), `0..1` (optional), * or `0..*` (unlimited zero or more), `1..*` (mandatory one or more).

Shared Aggregation

A "has-a" whole-part relationship indicated by a plain line with a hollow diamond at the whole end. Part instances can survive independently if the whole is destroyed.

Composite Aggregation

A strong ownership whole-part relationship indicated by a solid filled diamond. Part instances live and die with the parent container.

5. SUMMARY OF NINE OFFICIAL UML DIAGRAMS

UML organizes modeling tools into structural and behavioral categories:

- **Structural Diagrams (Static):** Class Diagram, Object Diagram, Component Diagram, Deployment Diagram.
- **Behavioral Diagrams (Dynamic):** Use Case Diagram, Sequence Diagram, Collaboration Diagram, Statechart Diagram, Activity Diagram.